

Old Age and the Decline in Financial Literacy

Abstract

We investigate whether knowledge of basic concepts essential to effective financial choice declines after age 60. Consistent with prior studies of cognitive decline in old age, we find that financial literacy scores decline by about one percentage point each year after age 60. We test for possible cohort effects and find that the rate of decline in financial literacy is nearly identical among men, stockowners, older, and college-educated respondents. Confidence in financial decision making abilities does not decline with age. Older respondents with lower financial literacy scores are more likely to pay a high mortgage interest rate and less likely to capture credit card rebates. A separate analysis using data that include measures of cognitive ability and financial literacy suggest that a natural decline in both fluid and crystallized intelligence in old age contributes to falling financial literacy scores.

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1. Introduction

Households age 60 and over hold 51% of all financial wealth in the United States¹. The proportion of U.S. households over the age of 60 is predicted to increase as the baby boom cohort and greater longevity contribute to population aging (Day, 2010). The transition to defined contribution plans tasks older Americans with greater responsibility for managing their own retirement assets and employing distribution strategies (Butrica, Iams, Smith, and Toder, 2009). Despite the importance of sound financial decision making among older Americans, little is known about either the magnitude of financial literacy decline in old age or about possible cognitive drivers of decision making quality.

There is evidence that financial decision-making ability declines in old age. Agarwal, Driscoll, Gabaix and Laibson (2009) show that the quality of credit decisions among borrowers erodes after peaking in the mid-50s. Despite evidence of improved general investment skill with age, investment performance declines significantly after age 70 (Korniotis and Kumar, 2011). Decision-making skills closely related to financial literacy such as the reliance on decision rules and resistance to framing decline in old age (de Bruin, Parker and Fischhoff, 2012). Hibbard, Slovic, Peters, Finucane and Tusler (2001) find that Medicare beneficiaries are nearly three times more likely than younger subjects to make errors when interpreting health plan information despite having more experience.

The observed decline in financial decision-making quality may be related to gradual mild cognitive impairment that occurs in old age. Boyle et al. (2012) find that the rate of cognitive decline in a sample of older adults is a significant predictor of incorrect responses to a financial decision-making test and increases susceptibility to financial scams. Older financial decision makers who experience a sharper decline in cognition report an increased difficulty in managing their money (Hsu and Willis, 2013). Cognitive ability, and in particular mathematical skills of the primary financial decision maker, is a strong predictor of the ability to avoid depleting net worth in later life (Smith, McArdle and Willis, 2011) and in making fewer financial mistakes (Agarwal and Mazumder, 2013).

Horn and Cattell (1967) attribute the declining performance on mathematical or spatial reasoning tasks after young adulthood and improved performance on tasks that require experience and knowledge to the theory of fluid and crystallized intelligence. Large cross-sectional analyses of performance on decision making tests such as word recall are indeed highest for respondents in their 20s and decline gradually through middle age before falling sharply after age 60, but scores on vocabulary tasks such as the ability to produce a synonym peak late in life and begin to fall after age 60 (Salthouse, 2009). Fluid intelligence decline appears to be the result of a general slowing in cognitive processing ability (Bugg,

¹ Calculated from the 2008 Survey of Consumer Finances.

Zook, DeLosh, Davalos and Davis, 2006) that has been linked to physiological changes such as a decline in frontal lobe volume after age 50 (DeCarli et al., 2005, Rushton and Ankney, 2009).

Financial literacy can be defined as the ability to understand fundamental financial concepts needed to make effective decisions. It requires an understanding of terminology, for example, a deductible on an insurance policy or the characteristics of a mutual fund, and the ability to comprehend how a higher deductible lowers an insurance premium or how greater diversification is a benefit of a mutual fund. Studies on information retrieval indicate that the ability to recognize terms may not decline in old age, but there is evidence that interpretation and general problem solving capability deteriorates. For example, Salthouse (2010) finds no significant decline in subjects' ability to solve crossword puzzles after age 60. Performance on more complex tasks that require the ability to retrieve information and use that information to solve a problem appears to worsen in old age. The ability to complete everyday tasks, for example to read and interpret instructions on a medicine bottle or interpret a rate chart on a telephone bill, decreases after age 60 (Diehl, Willis and Schaie, 1995). de Bruin et al.(2012) find that decreasing scores on fluid cognitive ability tests in old age contributes to the decline in performance on decisions that require both problem solving and accumulated knowledge. The ability to retrieve financial terms may or may not decline with age, but the ability to make an appropriate financial choice is particularly vulnerable to the age-related decline in reasoning skills.

Previous studies suggest a possible decline in financial literacy in old age. Lusardi, Mitchell and Curto (2014) identify differences in financial sophistication among older respondents in a subsample of the 2008 Health and Retirement Study. While the authors focus on how demographics impact knowledge scores, they note that respondents over 75 are less likely to understand basic investment concepts such as stock diversification and the importance of mutual fund fees. Consistent with a loss in fluid intelligence, respondents over 75 also score lower on numeracy questions. Lusardi, et al.(2014) do not investigate the rate of decline during old age nor does it investigate whether the decline is related to demographic differences among older cohorts. Descriptive results from van Rooij, Lusardi, and Alessie (2011) show that the proportion of respondents in the highest basic financial literacy quartile peaks in the 41-50 age category and is lowest among those ages 71 and older. The inverted U-shaped relation between age category and literacy is unexpected because rates of stock ownership and net worth are both related to higher financial literacy scores and rise with age. In multivariate analyses, van Rooij et al. (2011) do not segment respondents over the age of 60 and do not find that age is a significant predictor of financial knowledge.

Higher financial literacy scores have been linked to higher quality financial decisions. Hilgert, Hogarth and Beverly (2003) find that higher financial literacy scores predict on-time credit repayment, investment diversification, and mortgage refinancing. Lower financial literacy is associated with

incurring fees that are 50% higher on credit cards, particularly fees that require a more sophisticated awareness of credit terms (Lusardi and Tufano, 2009). A particularly costly financial mistake for older households in a falling interest rate environment is the failure to refinance a mortgage. Campbell (2006) finds that characteristics associated with financial sophistication predict wealth-maximizing refinancing behavior.

Our study adds to the existing literature on financial decision making and age by estimating the actual rate of decline in financial knowledge and measuring the decline within subgroups to address possible cohort biases. We also estimate the relation between the decline in fluid and crystallized intelligence and financial literacy scores, and investigate whether age-related financial literacy declines can affect financial decision-making quality. Studies that provide evidence of a decline in investment performance with advanced age (Korniotis and Kumar, 2011) and in credit decision making (Agarwal et al., 2009) do not directly estimate the decline in financial literacy that may be driving reduced performance in decision making ability. These performance studies also do not use data that allow them to accurately control for important demographic characteristics such as gender, race, and education that may influence observed behavior through differences in mortality rates or cohort differences in human capital investment. We extend the analysis of Lusardi et al.(2014) by using a much larger population of older households that allows subgroup analyses to address cohort and gender biases, and we model the rate of decline in financial literacy with advanced age.

We use a new financial literacy assessment instrument inserted into the Consumer Finance Monthly (CFM), a nationally representative monthly survey of credit behaviors, and obtain a sample of 3,873 respondents over age 60. The financial literacy score is composed of four questions, each within the topic areas of basic financial concepts, insurance, investments and credit knowledge. This unique instrument allows us to assess total financial literacy as well as more specific knowledge in financial topic areas. Financial literacy may be lower among older cohorts because of less investing experience in the pre-401(k) era, because women generally outlive men and may have delegated financial decisions to husbands, or because levels of higher education were lower for older cohorts. We estimate performance among domains and focus on insurance knowledge because rates of insurance ownership are higher among older cohorts. Our large sample size allows us to estimate the decline in financial literacy scores among college graduates, stockowners, men, and cohorts who reached age 60 before the rise in 401(k) popularity.

The survey also includes a self-assessed measure of confidence within each of the four financial literacy domains, and we estimate whether confidence declines with age and whether those with lower financial literacy have lower confidence. In order to identify possible cohort knowledge biases among topic areas, we estimate whether financial literacy declines with age for each of the 16 questions. Using

available financial decision quality variables, we estimate whether financial literacy scores predict interest rates on mortgages, observed understanding of financial questions, and the use of reward credit cards. Using the Health and Retirement Study, we create a financial literacy instrument and model scores as a function of age and measures of both fluid and crystallized intelligence to test whether age-related changes in cognition are associated with a decline in financial literacy scores.

We find a consistent linear decline in financial literacy scores after age 60 and a monotonic decline in scores among 5-year cohorts in the CFM data. Performance on all 16 questions declines significantly with age. The annual rate of decline in financial literacy scores is significant and similar among all subgroup analyses of men, stockowners, those with a college degree, and respondents who were 60 or older by 1992. The magnitude of age-related decline in scores is nearly identical among all four financial topic areas, including insurance. Confidence in one's ability to make financial decisions does not decline in old age and increases significantly for insurance. Age is positively related to financial overconfidence measured as the distance between objective literacy score and subjective financial confidence. Financial literacy scores predict interest rates on mortgages, observed understanding of financial questions, and the use of reward credit cards. Using data from the Health and Retirement study, we find that lower financial literacy scores within older age cohorts are related to the decline in fluid and in crystallized intelligence.

2. Methods

2.1. Financial Literacy Assessment

Some studies have measured financial literacy subjectively (e.g., Stango and Zinman, 2014) by asking respondents to estimate their perceived level of financial knowledge and capability. Other studies use measures that primarily measure numerical skills (Lusardi and Mitchell, 2007a, 2007b, Gustman, Stenmeier, and Tabatabai, 2012). Numeracy, or math capability, does not necessarily equate with having made the investment in knowledge of financial concepts needed to make optimal financial decisions (Almenberg and Widmark, 2011; Banks, 2010; Hung, Parker and Yoong, 2009). Other financial literacy measures focus on knowledge by asking respondents to recognize the correct definition of a financial term (Moore, 2003; Hilgert et al., 2003) while others attempt to incorporate skill by including items that require respondents to appropriately apply their knowledge (Mandell, 2008; Lusardi and Tufano, 2009).

Even among the measures that include financial knowledge and skill, none comprehensively cover the four main topic areas of financial literacy (Huston, 2010). Financial knowledge and skills encompass a broad array of topics ranging from awareness of financial terms to an understanding of financial concepts and the ability to select appropriate financial instruments. The objective of the financial literacy instrument used in this study is to measure both financial knowledge and the ability to apply

knowledge effectively. A research team initiated development of a financial literacy instrument, tested 89 potential questions and analyzed responses with the goal of choosing questions that were not biased in terms of age, gender, race and socioeconomic status, had an unambiguous correct response, and correlated well with other high quality questions. An eight-member panel of national experts in financial literacy and its assessment reviewed the project (including goals, design, model, instrument, scoring, and results) in order to assess the proposed methods and assessment instrument.

The final instrument of 20 items is selected from the best performing financial literacy questions according to reliability and validity statistics and the recommendations of the expert panel². In December, 2009 through 2013, the 20-item survey was included as a module in the Consumer Finance Monthly Survey conducted by the Center for Human Resource Research at The Ohio State University. The Consumer Finance Monthly collects demographic and detailed credit use information through a random digit dialing phone survey in the United States. The survey began in 2005 and includes over 10,000 completed financial literacy assessment instruments.

The sample in this study includes financial literacy responses from 3,873 respondents age 60 and older during this time period. The financial literacy instrument contains 20 items (Appendix B) covering the four content areas of personal finance – basics, investments, credit and insurance. Analyses show the questions within the instrument show high internal consistency³. Within each of those four personal finance content areas, there are two knowledge questions, two ability questions, and one confidence question. There are 16 questions used to measure the objective financial literacy score and four questions that measure confidence. Basic personal finance concepts include elements such as time value of money, purchasing power, and personal finance accounting. Intertemporal transfers of resources include both borrowing (bringing future resources into the present for consumption through the use of revolving credit and installment loans) and investment (saving present resources for future consumption through the use of savings accounts and investing through stocks, bonds, or mutual funds). Insurance questions include insurance instruments and risk management techniques. Financial literacy score is estimated as the percent correct out of 16 questions or out of four questions when scores are calculated within each topic area (basics, borrowing, investing and insurance). Decision making confidence is measured on a scale of 1 to 10 in each topic area and in total from 4 to 40 for the four topic areas.

The individual financial literacy questions require an understanding of basic financial products and an ability to apply them appropriately. In this sense, they test both knowledge of financial products

² The complete financial literacy assessment instrument can be found at <https://sites.google.com/site/pfinttu/flat>

³ Construct validity estimates for our financial literacy assessment instrument are higher than for previous financial literacy instruments, and the sample size is more than twice as large as any previous literacy module (Hung et al., 2009). Cronbach Alpha estimates and correlations among questions can be found in Appendix A.

that will likely improve from age and experience, and some reasoning skill which may decline in advanced age. For example, an insurance question asks what impact a higher deductible will have on an insurance premium. This requires an understanding of the financial terms deductible and premium, and the ability to think through how a higher insurance deductible will affect the cost of insurance.

2.2. Measuring the Decline in Financial Literacy in Old Age

Our first objective is to test whether financial literacy scores decline among respondents age 60 and older, and to estimate how the rate of decline changes in advanced age. We then test whether respondent characteristics other than age are associated with financial literacy. The greatest challenge to any cross sectional analysis of knowledge assessment is the possibility of cohort effects that may create estimation biases. We conduct a number of subgroup analyses in order to test whether the hypothesized negative relation between age and financial literacy remains consistent.

To create Table 1, we calculate average financial literacy score for each year of age among the 3,873 respondents age 60 through 94 in the CFM for a total of 35 years. For example, there are 265 respondents age 60 and the average financial literacy score is 61% among 60-year old respondents. Cross sectional estimation of the marginal change in average test score for each year of age is common in the cognitive aging literature (Salthouse, 2010)⁴. We model average score as the dependent variable and age as a single independent variable. Regressions are estimated using the average overall 16-question financial literacy score as a dependent variable, the overall 4-question confidence score, scores within each of the 4-question topic areas (basics, borrowing, investments and insurance), and confidence within each of the 4-question topic areas. Average financial literacy score for all ages is included in Figure 1, and Figure 2 shows the average decline in financial literacy and average confidence score by age. Figure 3 shows average score within each topic by age.

2.2.1. Controlling for Respondent Characteristics

In the multiple regression analyses, we model financial literacy as a function of demand for financial human capital. The decision to incur the direct and indirect costs of attaining financial

4 Although there is some debate about the use of cross-sectional data to identify age-related decline in task ability, Salthouse (2009) illustrates how longitudinal estimate biases caused by subject learning create significant problems in panel data. Even questions that measure spatial orientation and word recall show increased ability in subsequent panel surveys among young and old respondents, while cross-sectional data show a consistent decline (Salthouse, 2009). The best method of estimating age-related task decline is to carefully reduce potential cohort effects through empirical models that control for differences among age groups that may be related to task ability.

knowledge is a function of the time and transaction costs and the discounted expected utility from making more effective financial decisions in future periods.

Higher education may proxy a lower cost of information acquisition, a lower rate of time preference, or may involve direct exposure to financial information via business or economics coursework – all of which will increase expected financial literacy. Home ownership may be related to financial literacy both through experience with related financial products (for example insurance concepts), and by increasing the expected return to investment in tax rules. Likewise, stock ownership may involve a fixed information cost that suggests a greater expected benefit from investment in financial human capital (Peress, 2004). The use of tax-sheltered accounts requires an initial financial human capital investment and may help explain greater investment knowledge among those who actively saved during the 1980s and 1990s when the use of sheltering instruments expanded in the U.S. We use a question that asks respondents whether they have “any money in tax advantaged accounts including IRAs, Keogh plans, variable annuities or 529 plans” or “money in retirement plans through former employers such as a 401(k) or 403(b).”

Financial wealth will increase the expected future payout from investing time and effort into making more informed financial decisions (Peress, 2004). We use the top income and wealth quintile to capture the incentive to invest in financial information among those with the most money to manage. To some extent, home ownership, stock ownership, the ownership of tax sheltered accounts and marital status will also capture financial resource availability. Racial differences in financial literacy may be attributable to differences in financial human capital inherited from parents or to differences in the frequency of financial knowledge transfer in social interactions (Brown, Izkovic, Smith and Weisbenner, 2008). Women may have lower financial literacy if households allocate financial decisions to the spouse with a lower relative cost of financial capital acquisition (Croson and Gneezy, 2009). Smith et al.(2010) find that women are less likely to be the primary household financial decision maker in older household cohorts. We also include dummy variables for the year and month of the survey.

In order to estimate the impact of age on financial literacy among individuals, we model the percent correct on financial literacy topic areas and total score (percent correct of 16 questions) as a function of age and control variables using an ordinary least squares (OLS) regression:

$$FinLit_i = a + bAGE_i + \gamma X + \varepsilon_i \quad (1)$$

where $FinLit_i$ is the objective financial literacy score (percent correct out of 16 questions) for respondent i and AGE is the respondent's age in years. We include a matrix of control variables (X) that yield a vector of coefficient estimates (γ). The control variables include education, income, and wealth levels, along with gender, race, marital status, and ownership status of home, tax shelters, and stocks (see equation 1).

In a second regression (see equation 2), interaction variables of age measured as a continuous variable (*AGE*) multiplied by whether the respondent is age 60-69, 70-79, or 80 years of age or older (*AGECAT*) are substituted for a linear age variable to account for possible slope differences in the effect of age on financial literacy.

$$FinLit_i = a + bAGE_i * AGECAT + \gamma X + \varepsilon_i \quad (2)$$

To provide a comparison of the independent age effect across the entire sample, we model total financial literacy score as a function of age (*AGE*) and age-squared (*AGE*²) to capture the hypothesized inflection point of financial literacy in middle age (see equation 3).

$$FinLit_i = a + bAGE_i + cAGE_i^2 + \gamma X + \varepsilon_i \quad (3)$$

We also specify age using 5-year age cohorts using age 45-49 as the reference category, and provide actual average financial literacy scores by age, to create Figure 1. For example, the age cohort scores are estimated based on the coefficients with control variables set to their mean value from the model:

$$FinLit_i = a + bAGE_{25-29}_i + cAGE_{30-34}_i + dAGE_{35-39}_i + eAGE_{40-44}_i + fAGE_{50-54}_i + gAGE_{55-59}_i + hAGE_{60-64}_i + jAGE_{65-69}_i + kAGE_{70-74}_i + lAGE_{75-79}_i + mAGE_{80-84}_i + nAGE_{85-89}_i + oAGE_{90-94}_i + \gamma X + \varepsilon_i \quad (4)$$

2.2.2. Addressing Sample Biases

Differences in experiences or incentives to invest in financial knowledge may affect performance on the financial literacy test. To minimize possible biases in older cohorts, we create subsamples that reduce the most significant sources of age-related financial knowledge variation that are unrelated to cognitive decline.

Because rates of educational attainment have risen in the United States during the 20th century (Day, 2010), younger cohorts may be more likely to have taken an economics or finance course in college. To reduce the potential bias caused by lower educational attainment by older cohorts, we estimate our model only on older respondents who have a college or graduate school education. Men, particularly in older cohorts, may choose to invest in financial knowledge as a result of specialized labor in household production. Because longevity is higher among women than men, our results may be biased by a larger proportion of older, less knowledgeable women. To correct for gender-related knowledge differences, we estimate the model among males only. Cohort differences in rates of return on stock investments may drive variation in equity market participation (Malmendier and Nagel, 2011). If households over 60 are less inclined to invest in equities due to their poor performance in the 1970s, this

may have influenced the decision to seek out investment information about stocks. To address the bias or reduced preference for risky asset ownership, we estimate the model using households who directly hold stock or mutual fund investments. Finally, since financial literacy is strongly related to stock market participation (van Rooij, Lusardi and Alessie, 2011), and household stock ownership rates peak among households age 45-54 and rose significantly between 1983 and 1992 in qualified retirement plans (Poterba and Samwick, 1995), we estimate our model only for households who were age 60 or older in 1992.

Another potential criticism of estimating the relation between age and financial literacy is the possibility that older households were less likely to be exposed to financial instruments less common in their peak borrowing and saving life cycle years. One exception is insurance products. Cohort ownership rates of life insurance are higher among older households than among the baby boomer cohort (Chen, Wong and Lee, 2003). In addition, insurance products were a common sheltered savings vehicle prior to the 401(k) and IRA era that began in the 1980s. The four financial literacy questions related to household insurance present less potential cohort bias than other topic areas. We estimate OLS regressions on the percent correct (out of four questions) using equation 1 from within each of the four financial literacy topic areas, including insurance, investments, borrowing and basics in order to detect possible differences in the marginal effect of aging against knowledge in different literacy domains.

We estimate sixteen separate logistic regressions using equation 1 on each financial literacy question to determine whether the results are driven by a subset of questions that may be age or cohort biased. We separately estimate standardized beta coefficients in order to calculate the relative independent strength of the age variable among independent predictors of financial literacy within the multivariate model.

2.3. Financial Confidence

The financial literacy assessment instrument includes four questions that ask the respondent to assess how confident they are at making financial decisions within each of the topic areas. It is possible that older subjects are not aware of declines in their financial decision making ability and may or may not remain confident of their financial capabilities. Understanding whether actual ability and age are related to higher confidence is important in understanding whether seniors are potentially vulnerable to decision making mistakes from overestimating their decision making ability. In order to better understand an overconfidence in one's abilities, we investigate the characteristics that predict a high level of confidence among respondents with a low level of knowledge.

Confidence in financial ability is measured through a question asking respondents to rate on a scale of 1 to 10 how confident they are in making decisions within each of the four financial literacy topic areas. We use an OLS model to estimate predictors of confidence in each topic area, and the summed

total of all four topic areas, as a function of age, financial literacy within that topic area (or total financial literacy score), and household characteristic control variables:

$$Confidence_i = a + bFinScore_i + cAGE_i + \gamma X + \varepsilon_i \quad (5)$$

where $Confidence_i$ is either confidence in a specific topic area (basics, borrowing, investment, insurance) or total confidence (percentage total of all four topic areas) for respondent i (see equation 5). Confidence in each topic area is respondent-assessed on a scale of 0 (no confidence) to 10 (highest confidence).

$FinScore_i$ is the specific topic area objective score (percent correct out of 4 questions) or the total score ($FinLit$) for respondent i . Coefficient estimates indicate the marginal effect of age on financial confidence controlling for actual ability. The sample size decreases slightly to 3,403 because of some missing responses among those who completed the financial literacy test.

Respondents whose confidence score is an average of at least 8 for all four topic areas (or a total score of at least 80%), and whose objective financial literacy score is in the lowest quartile, are considered overconfident. Since slightly higher confidence may improve financial outcomes for those with adequate financial literacy, we choose to specify overconfidence as very low financial literacy and very high confidence. We select a logistic model (see equation 6) of overconfidence as a function of age (AGE), and interaction of age and financial literacy ($AGE*FinLit$), and the same matrix of control variables (X).

$$Overconfidence_i = a + bAGE_i + cAGE*FinLit_i + \gamma X + \varepsilon_i \quad (6)$$

2.4. Financial Literacy and Decision-Making

Prior research shows a decline in financial decision-making quality in old age. We examine whether lower financial literacy is associated with an increased likelihood of selecting more efficient financial products and of understanding the questions posed in the survey. We also investigate whether overconfidence, independent of a lower overall financial literacy score, increases the likelihood of poor financial decision-making. In order to understand whether the negative impact of lower financial literacy scores and higher overconfidence differs among age cohorts, we include interaction effects of each variable with older age categories.

To gain an understanding of decision-making quality of older households, we create logistic regression models that include age categories, financial literacy, financial confidence and the age-score interactions, while including the matrix of control variables (see equation 7). We select two financial outcomes available in the CFM as the most unambiguous measures of decision-making quality among

older households. The first outcome measures whether respondents with a mortgage have an interest rate higher than what is normatively optimal. Agarwal, Driscoll, and Laibson (2013) find that optimal refinancing differentials between existing mortgage and current interest rates are never greater than 200 basis points. We set a threshold of 6.5%, which is the lowest rate among respondents with a mortgage who fall in the highest rate quintile and roughly 200 basis points above the average rate during the survey period (4.46%). Campbell (2006) finds that variables related to financial literacy such as formal education are a significant predictor of owning a mortgage 2% above market rates. The second outcome measures whether the respondent is a convenience credit card user and does not take advantage of costless interchange fee reimbursement through a reward card (Ching and Hayashi, 2010). We censor to only respondents who own a credit card and who indicate that they pay off their credit card balance in its entirety each month, and thus do not consider card interest rate a salient characteristic. Retailers pay \$30 billion in credit card interchange fees at a cost of \$350 per household, and 44% of this (or \$150 per household) amount is rebated to consumers through reward cards (Ching and Hayashi, 2010). The failure to capture this rebate among convenience users represents a significant loss arising from low financial literacy. The third outcome we model is whether the interviewer believes the respondent had a good understanding of the survey questions to test whether financial literacy predicts a global subjective assessment of financial decision making ability.

$$Outcome_i = a + bAGE70-79_i + cAGE80Plus_i + dFinLit_i + eOC_i + fFinLit*AGE70-79_i + gFinLit*AGE80Plus_i + hOC*AGE70-79_i + jOC*AGE80Plus_i + \gamma X + \varepsilon_i \quad (7)$$

where $Outcome_i$ is either the mortgage rate, credit reward, or interviewer's assessment status for respondent i . $FinLit_i$ represents the financial literacy score and OC_i represents the overconfidence status (1=overconfident; otherwise 0) for respondent i .

2.5. Financial Literacy Decline and Cognitive Aging

If the decline in financial literacy in old age is not explained by cohort differences or household characteristics, it is possible that cognitive decline in advanced age is responsible for reduced financial decision-making ability. The 2010 Health and Retirement Study (HRS) contains a module that asks financial literacy questions to a subsample of respondents. The HRS also asks respondents to answer questions in a cognition module that include established measures of fluid and crystallized intelligence. Through the use of cognition measures, we are able to test whether lower financial literacy scores are related to by cognitive aging.

Financial literacy questions are drawn from an HRS module of questions that assess respondent "financial sophistication and investment decision making." The HRS module includes questions that

measure financial literacy and other concepts related to financial awareness. We select the questions from the module that measure financial literacy. These include the original three questions from Lusardi and Mitchell (2007a) and four additional questions that have a specific answer (“buying a single company stock usually provides a safer return than a stock mutual fund”), and we avoid questions with no specific correct response (“are you considering investing in the stock market for the next year?”). The financial literacy instrument is an additive score that includes one question on interest compounding and one on inflation (each of these may be related to numeracy), whether a single stock is safer than a mutual fund, whether stocks historically provide higher returns than bonds or savings accounts, whether an employee should have a lot of their retirement savings in employer stock, whether foreign stocks should be avoided, and whether bond values are inversely related to interest rates.

Using the percent correct from an additive score of 7 financial literacy-related questions as the dependent variable, we model using OLS financial literacy as a function of age and cognitive function while controlling for a matrix of household characteristics (X) for respondents age 60 and older (see equation 8). We estimate models that specify age as a linear variable and as 5-year cohorts using age 60-64 as the reference category (see equation 8).

$$HRS-Score_i = a + bAGE_i + cRecall_i + dVocab_i + \gamma X + \varepsilon_i \quad (8)$$

where $HRS-Score_i$ is the percent correct out of 7 financial literacy-related questions from the HRS for respondent i and AGE is the respondent’s age in years (or age category using 5-year intervals). We calculate fluid intelligence using a combination of immediate and delayed word recall scores, a reliable measure available in the Health and Retirement Study (HRS) (McArdle, Fisher and Kadlec, 2007). $Recall_i$ is the word recall is the number of nouns (out of 10) recalled by respondent i . $Vocab_i$ is a vocabulary measure adapted from the WAIS-R crystallized intelligence test that asks respondents to define each of five words (such as plagiarize or perimeter). Responses are scored between 0 (wrong) and 2 (perfectly correct) for a total score of between 0 and 10. Analyses are performed using a sample of 1,109 respondents who answered the financial literacy module for the analyses without the cognition variables and a total of 887 respondents who completed both the cognition and financial literacy questions.

3. Results

3.1. Financial Literacy Decline in Old Age

Figure 1 shows the average financial literacy scores by age and the score predicted by the multivariate models that specify age as a quadratic variable and as 5-year cohorts using the full CFM sample. Actual and predicted scores using the two function forms show a clear concave relation between

actual financial literacy score and predicted score controlling for other respondent household characteristics. Average financial literacy scores within each year of age increase up to roughly age 50, while the quadratic specification controlling for demographic characteristics peaks at age 49, and 5-year cohort model is highest in the age 40-44 group (although it is not statistically different from the 45-49 year old reference group). Within the cohort model, the first cohort to have scores statistically lower than the 45-49 year old reference group is age 65-69 (3% lower) and predicted financial literacy scores decline at a rate of between 5 and 7 percentage points for each 5-year later life cohort. The predicted financial literacy percentage score (when compared to respondents age 45-49) is roughly the same among respondents between age 25 and 29 (7.9% lower) as it is for respondents age 70-74 (8.3% lower) and falls to 35% lower for respondents age 90 or older.

Univariate regression analyses in Table 1 model the yearly change in average financial literacy scores among respondents between the age of 60 and 94. Regression results show that age is a strong and consistent predictor of financial literacy score. With each year of age after 60, the average score falls by 1.5 percentage points, and the relation is consistent (R-square of 0.96). The relation between age and average financial literacy score is consistent among the four decision making topic areas ranging from a 1.42% decrease each year within basic questions to 1.65% for investment questions. Although financial literacy scores decline with age, confidence in financial decision making does not. Confidence in financial decision-making ability increases slightly with age, but the relation is statistically significant only within the insurance domain.

Figure 2 shows the similar rate of decline in average financial literacy score in the CFM and in average word recall ability by age in the HRS. Although episodic memory and financial literacy score decline at roughly the same rate after age 60, confidence in financial decision making ability remains relatively unchanged with age. The percentage of overconfident respondents with high self-assessed ability and low objective literacy scores increase from about 10% in the 60s to higher than 30% among respondents over 85. Figure 3 illustrates the consistent decline in financial literacy score with age among all four topic areas.

Sample characteristics in Table 2 indicate consistently lower financial literacy scores in all topic areas among respondents age 70-79 (49%) and 80+ (32%) than among respondents age 60-69 (62%). Older respondents have lower average financial literacy scores in all topic areas including insurance. Scores are much higher among respondents with a college (61%) and graduate (66%) degree than among respondents with a high school (39%) or below high school (25%) education. Scores are higher among whites, men, homeowners, those who are married, stock owners, and increase monotonically with wealth and income quintile. Greater financial resource availability is associated with higher financial literacy scores. Older households are slightly more confident in their financial decision making abilities (Table

2b). Since average financial literacy score declines with age, it is not surprising that a higher percentage of respondents age 80 or above (19.3%) are overconfident than respondents age 70-80 (10.4%) and age 60-69 (4.7%). Financial confidence is only slightly higher among more educated respondents, but a much higher percentage of respondents with a high school or below high school education are overconfident.

Table 3 presents regression results that estimate the financial literacy score by individuals with and without control variables⁵ among respondents 60 years and older in the CFM. Unlike Table 1, the regression model predicts individual financial literacy score for each of the 3,873 respondents as a function of the respondent's age (rather than average financial literacy score for all respondents that have the same age in Table 1). Because financial literacy will vary among respondents of the same age with different demographics and life experience, the fit of the relation between age and financial literacy is lower (R-square 0.17) but still statistically significant. Each year of age is associated with a 1.36 percentage point decline in the total financial literacy score. When age is sorted into 5-year groups, respondents age 70-74 have significantly lower financial literacy scores than respondents age 60-64. The predicted financial literacy score falls by between 6 and 9 percentage points with each older cohort group. Respondents age 90 or older score 41.7 percentage points lower on average than respondents age 60-64. The age effect declines slightly to 1.02 percentage points per year when control variables are included in the model, but the effect is no less consistent. The model with control variables explains 37% of the variation in observed financial literacy. Coefficients in the multivariate model also decline slightly in magnitude but remain significant when age is measured in 5-year groups and the fit is identical to the linear age model.

3.1.1. Addressing Sample Biases

Regressions in Table 4 model financial literacy score using the linear age variable within subsamples in order to correct for possible biases in experience or motivation to acquire financial knowledge. Among households with a college education, the magnitude of the decline in financial literacy score is slightly higher than in the full sample (1.10 percentage points per year). The annual decline in financial literacy is similar in a sample of men (0.96 percentage points per year) compared to a full sample of male and female respondents. Among stock or mutual fund owners, the annual decline in financial literacy is 0.98 percentage points and also statistically significant. Among the cohort of respondents who were age 60 or older in 1992, the magnitude of decline with age is 1.37 percentage points per year. The estimated linear decline in financial literacy is similar among subgroups.

⁵ A quadratic age specification yields an insignificant age-squared coefficient that is near zero indicating no inflection point of age on financial literacy in the older sample.

Financial literacy regression results for individual topic areas in Table 5 show the marginal impact of age on predicted score within the four areas of financial literacy knowledge. The annual decline in financial literacy scores is consistent among all four topic areas, and the magnitude of the effect is comparable (ranging from a 0.94 percentage point decline per year for basics to a 1.10 percentage point decline in borrowing knowledge). All results are statistically significant. Insurance knowledge, which would increase with age if results are driven by cohort financial instrument familiarity effects, declines with age at roughly the same rate (0.96 per year) as basic financial knowledge. The consistency of the age decline among financial literacy topic areas can also be seen when we model the correct response to individual questions. Table 6 shows that the likelihood of providing a correct response to each financial literacy question declines significantly with age. Of the 12 control variables (including important human capital-related characteristics such as education, income race, gender), age is the strongest independent predictor of providing the correct answer for 11 of the 16 financial literacy questions. Of the remaining questions, age is the second and third strongest predictor. Interestingly, the weakest age effect is observed for a question that asks about appropriate mortgage types for a first-time home buyer. The strongest age effect occurs for the questions on the deductibility of interest and the use of money market accounts.

3.2. Financial Confidence and Decision-Making

Table 7 shows multivariate analyses of confidence in managing money, managing credit and debt, using investment products and using insurance, as well as overconfidence measured as the difference between self-assessed and actual financial literacy. Confidence in overall financial decision making ability increases with age, and also within all topic areas. More financially literate respondents are also more confident for each topic area except insurance. Respondents who are less knowledgeable about insurance are not less confident about their insurance knowledge. Older respondents are more likely to be confident about their ability to make insurance and investment decisions.

The likelihood of being overconfident with one's financial knowledge increases with age. Each year of age after 60 increases the likelihood of having high confidence and low financial literacy scores by 7 percent. Higher levels of education are associated with a much lower likelihood of overconfidence, as are being male and white.

The only variable that consistently predicts confidence in all four areas is homeownership. Although age is related to increased financial confidence, the multivariate models explain little variation in financial confidence and the marginal effect of age is weak compared to other variables (the effect of an additional 20 years of age is roughly equal to homeownership in predicting total confidence).

Estimation of the impact of age and financial literacy on decision quality outcomes available in the CFM are presented in Table 8. Respondents with a higher financial literacy score are more likely to have a competitive mortgage rate beneath a threshold at which a borrower should refinance. There is no independent relation between overconfidence and a low mortgage rate. Like Campbell (2006), we find that education and wealth are significant predictors of a low mortgage rate. More financially literate and older convenience credit card users were also more likely to have a reward card. Again, there is no relation between overconfidence and reward card ownership. More financially literate respondents were more likely to have a good understanding of the survey questions according to the interviewer. Age and overconfidence had no independent impact on understanding.

3.3. Financial Literacy Decline and Cognitive Aging

Table 9 shows results using data from the Health and Retirement Study that include measures of respondent fluid and crystallized intelligence as well as a different measure of financial literacy. Coefficients represent the percentage change in financial literacy score out of seven questions included in the HRS.

The HRS financial literacy questions show a consistent, but slightly weaker, decline in financial literacy among respondents 60 and older. The linear age specification is negative and statistically significant, and the 5-year age group coefficients are negative and monotonic but only reach the level of a statistically significant difference from age 60-64 by the 75-79 age group. Similar to the model using CFM data, controlling for household characteristics slightly reduces the linear age effect. The magnitude of financial literacy decline is similar among 5-year age groups who have reached at least age 80 after demographics are controlled.

Both the measures of crystallized and fluid intelligence are statistically significant and the coefficients are identical as predictors of higher financial literacy scores. Once cognitive ability is controlled, the age coefficient remains statistically significant but the magnitude of the annual decline falls by 41%. When age is specified using 5-year age groups, none of the age groups is statistically significant after we control for word recall and vocabulary ability. Again, we find that fluid and crystallized intelligence predict financial literacy scores by a similar magnitude.

4. Conclusion

Using a new financial literacy instrument and a large, nationally representative sample, this study is the first to directly measure the decline in financial literacy in advanced age. We find a consistent linear decline in average financial literacy score of about 1 percentage point per year among respondents over 60. In order to correct for a number of potential biases due to differences in experience and

incentives to invest in financial knowledge, we conduct a number of subsample analyses. We find that the effect of age on financial literacy score is strong and the magnitude of the effect is consistent across all models. The likelihood of correctly answering each of the 16 questions used in the literacy score declines significantly with age, and the rate of decline is not different within a topic area (insurance) that had higher rates of ownership among older cohorts. Confidence in financial decision-making ability does not decline with age, and the likelihood of having high confidence and a low financial literacy score rises in old age. Older respondents with higher financial literacy are more likely to make higher quality financial decisions.

In a separate analysis using the Health and Retirement Study that includes measures of cognition, we find that controlling for fluid intelligence (word recall) and crystallized intelligence (vocabulary ability) eliminates the statistical significance of age cohort categories as predictors of financial literacy and weakens the significance of a linear age variable. The results suggest that falling financial literacy scores in old age are related to the well-established decline in cognition documented in the aging literature. The results are similar to other studies of changes in financial preferences, such as reduced risk tolerance in old age, which disappear when cognitive capacities are included as predictor variables (Henninger, Madden and Huettel, 2010). The implication is that a decline in financial literacy may be a natural consequence of cognitive changes in old age.

Our results add to the literature on observed declines in financial performance with advanced age by providing evidence consistent with the decline in financial literacy being caused by a general decline in cognition. For example, Korniotis and Kumar (2011) show a decrease in investment performance that mirrors observed declines in cognitive ability by age. Our study shows that the decline in performance may be attributed directly to an age-related decrease in financial knowledge and the ability to apply knowledge correctly to financial decision-making. We are also able to better control for possible confounding effects closely related to financial knowledge by using household-level control variables such as education, homeownership, and race. For example, the proportion of individuals with a college degree declines among older cohorts, which could explain the decline in financial literacy and investment performance. However, we find that college-educated respondents over 60 experience a decline in financial literacy with age that is similar in magnitude to the full sample.

Multivariate analyses censored by financial literacy within four topic areas provide the most convincing evidence that our results are not driven by cohort effects. Life insurance ownership rates are higher among older age cohorts (Chen, Wong and Lee, 2003). Three of the four insurance questions test knowledge and application of life insurance concepts (the fourth tests knowledge of insurance deductibles). The magnitude of annual decline in scores on insurance knowledge (0.96%) is nearly identical to the decline in basic financial literacy (0.94% per year) and similar to the annual decline in

investment literacy (1.02%). We find a similar annual decline among stockowners (0.98%), evidence that cohort effects related to differences in equity market participation are not driving the decline in financial literacy. Both significantly lower financial literacy scores among the youngest respondents and a consistent linear decline in financial literacy after age 60 are consistent with crystallized intelligence theory rather than a technology-based cohort effect that favors younger respondents.

Empirical evidence from cross-sectional studies of cognitive aging show a steady linear decline in tests of fluid intelligence and a more modest decline in problem solving that involves both processing ability and information retrieval. Financial literacy questions in the Health and Retirement Study (HRS) assess numerical ability (for example the estimation of compound interest over time), as well as problem solving skills (such as whether an employee should own employer stock). Our results show that, unlike the ability to solve a crossword puzzle (Salthouse, 2010), financial literacy requires both the ability to recall terms and the ability to correctly solve problems that require fluid intelligence. Fluid intelligence measured through word retrieval and crystallized intelligence measured through a vocabulary test have an equal impact on predicted financial literacy scores in the HRS.

A decline in financial skills may not lead to poor financial outcomes if individuals recognize and anticipate the decline. For example, recognition of diminished investment skills may increase demand for annuitization or the delegation of important financial decisions to a trusted advisor. Studies of trading frequency provide some evidence that older investors are less overconfident than younger investors (Barber and Odean, 2001). In contrast, our study finds that, in aggregate and within all financial decision making domains, advanced age increases overconfidence in financial decision-making abilities. The largest marginal effects are within the investment and insurance topic areas. The less educated, non-whites, and females are more likely to be financially overconfident in the old age sample. We do not find, however, that individuals who are most overconfident are less likely to optimally refinance or to choose a reward credit.

Our results show that it is not so much the imbalance between confidence and knowledge that is causing poor financial decisions, but the low financial literacy itself. Among the aged within similar decision-making domains, there is an inclination to reject evidence of declining mental abilities. For example, older drivers generally do not perceive a decline in their driving skills despite a predictable deterioration in sensory ability with advanced age (Holland and Rabbitt, 1992). However, they report that those who did perceive a decline in their abilities, and those who took an objective test that provided evidence of a decline, modify their driving behavior to reduce the likelihood of getting into an accident. It is possible that increased awareness of the natural decline in cognitive abilities essential to making effective financial decisions will lead to greater demand for more passive financial instruments such as annuities or passive investment vehicles that automatically rebalance. It may also increase demand for

professional services such as financial planning, accounting and legal assistance that substitute for one's own decision-making ability. The simultaneous decline in financial literacy and increase in decision-making confidence with advanced age also has implications for national retirement policy. Programs (such as Social Security) that automatically annuitize retirement income and do not require a retiree to manage withdrawal and investment, may improve social welfare (Diamond, 2004).

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Table 1 **Univariate Relation between CFM Financial Literacy Score and Age**

Each row in this table presents the results of a univariate regression of the annual change in mean financial literacy score (percentage points--pp) for each age from 60 to 94 (dependent variable) on age (independent variable)

<i>Financial Literacy Measure</i>	<i>Annual Change (pp)</i>	<i>T-statistic</i>	<i>R-Square</i>
Overall Financial Literacy Score (16 Questions, %)	-1.53	26.27***	0.96
Overall Confidence in Literacy	0.11	1.86	0.07
Overconfidence in literacy	1.63	19.88***	0.92
Basic Literacy	-1.42	20.86***	0.93
Borrowing Literacy	-1.61	22.30***	0.94
Investment Literacy	-1.65	21.48***	0.94
Insurance Literacy	-1.43	18.75***	0.92
Confidence in Managing Money	0.06	1.12	0.01
Confidence in Credit	0.05	0.99	0.00
Confidence in Investing	-0.00	0.05	0.00
Confidence in Insurance	0.31	3.91***	0.31

***, **, * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively

Table 2a CFM Sample Financial Literacy Scores and Variable Frequency/Mean

Each panel in this table presents the total financial literacy score and topic area financial literacy scores for each variable, along with the sample frequency or mean for each variable attribute.

	Financial Literacy Objective Scores (0-100)						
	Total Score %/(std)	Basics %/(std)	Borrowing %/(std)	Investment %/(std)	Insurance %/(std)	Frequency Or Mean	N
Age							
Age 60-69	62 (26)	62 (31)	63 (30)	57 (34)	65 (31)	51.1%	1979
Age 70-79	49 (28)	48 (33)	51 (32)	43 (36)	54 (34)	31.0%	1201
Age 80 or above	32 (24)	34 (29)	33 (29)	25 (29)	37 (32)	17.9%	693
Education							
<High School	25 (20)	22 (25)	30 (28)	16 (22)	31 (30)	4.9%	190
High School	39 (26)	38 (30)	42 (32)	31 (32)	44 (33)	23.8%	922
Some College	51 (26)	51 (32)	54 (32)	44 (34)	55 (32)	25.9%	1003
College	61 (26)	62 (31)	61 (31)	58 (34)	64 (31)	24.5%	949
Graduate	66 (26)	67 (30)	64 (30)	62 (34)	69 (31)	20.9%	809
Race							
White	54 (28)	54 (33)	55 (32)	48 (36)	58 (33)	88.6%	3431
Non-White	42 (27)	40 (31)	47 (33)	35 (33)	47 (34)	11.4%	442
Gender							
Male	58 (28)	58 (33)	59 (32)	54 (36)	61 (33)	43.3%	1677
Female	48 (28)	48 (33)	50 (32)	41 (34)	53 (34)	56.7%	2196
Homeownership							
Homeowner	54 (28)	54 (33)	55 (32)	49 (36)	58 (33)	90.2%	3493
Not Homeowner	38 (28)	40 (33)	40 (32)	30 (32)	43 (33)	9.8%	380
Marital Status							
Married	59 (27)	59 (32)	60 (32)	54 (35)	62 (33)	54.6%	2115
Unmarried	45 (28)	44 (32)	47 (32)	38 (34)	50 (34)	45.4%	1758
Tax Sheltered Status							
Tax Sheltered	64 (26)	66 (30)	63 (31)	62 (34)	67 (30)	30.1%	1166
No Tax Sheltered	48 (28)	48 (33)	51 (32)	41 (35)	53 (34)	69.9%	2707
Stock Ownership							
Stock/MF Owner	64 (25)	64 (31)	63 (30)	61 (33)	67 (31)	43.1%	1669
No Stock/MF	45 (28)	45 (32)	48 (32)	37 (34)	50 (33)	56.9%	2204
Income Level							
Lowest Income	39 (27)	37 (31)	42 (32)	32 (33)	46 (33)	\$7,953	775
Quintile 2 Income	46 (27)	46 (32)	48 (32)	37 (34)	51 (33)	18,881	775
Quintile 3 Income	57 (25)	55 (31)	59 (29)	51 (34)	61 (31)	33,756	775
Quintile 4 Income	62 (24)	64 (30)	63 (29)	56 (33)	65 (31)	59,427	774

<i>Highest Income</i>	69 (22)	70 (27)	69 (28)	66 (32)	72 (27)	165,598	774
Wealth Level							
<i>Lowest Net Worth</i>	37 (27)	37 (32)	40 (33)	30 (32)	42 (34)	(\$11,418)	775
<i>Quintile 2 NW</i>	44 (27)	43 (32)	48 (32)	34 (33)	50 (33)	104,917	775
<i>Quintile 3 NW</i>	53 (27)	54 (33)	55 (32)	47 (35)	57 (33)	250,351	775
<i>Quintile 4 NW</i>	60 (25)	61 (29)	60 (30)	56 (33)	64 (31)	459,537	774
<i>Highest Net Worth</i>	68 (24)	68 (29)	67 (29)	67 (32)	71 (29)	1,797,815	774

Table 2b CFM Sample Financial Confidence Score and Overconfidence Status

Each panel in this table presents the financial confidence score and the percentage that are overconfident for each variable attribute.

Financial Confidence Scores (0-100) and Overconfidence Status			
	<i>Confidence Score %/(std)</i>	<i>Overconfidence %</i>	<i>N</i>
Age			
<i>Age 60-69</i>	72 (19)	5.7%	1979
<i>Age 70-79</i>	75 (19)	10.4%	1201
<i>Age 80 or above</i>	74 (21)	19.3%	693
Education			
<i><High School</i>	69 (25)	16.2%	190
<i>High School</i>	73 (21)	16.6%	922
<i>Some College</i>	72 (20)	8.7%	1003
<i>College</i>	75 (17)	6.5%	949
<i>Graduate</i>	74 (18)	4.9%	809
Race			
<i>White</i>	73 (19)	9.0%	3431
<i>Non-White</i>	70 (23)	14.2%	442
Gender			
<i>Male</i>	74 (19)	7.2%	1677
<i>Female</i>	73 (19)	11.5%	2196
Homeownership			
<i>Homeowner</i>	74 (19)	9.4%	3493
<i>Not Homeowner</i>	66 (22)	11.9%	380
Marital Status			
<i>Married</i>	74 (18)	7.4%	2115
<i>Unmarried</i>	72 (21)	12.2%	1758
Tax Sheltered Status			
<i>Tax Sheltered Acct.</i>	76 (16)	3.8%	1166
<i>No Tax Sheltered</i>	72 (20)	11.1%	2707

Stock Ownership			
<i>Stock/MF Owner</i>	77 (16)	5.0%	1669
<i>No Stock/MF</i>	72 (20)	10.7%	2204
Income Level			
<i>Lowest Income</i>	69 (23)	13.1%	775
<i>Quintile 2 Income</i>	70 (22)	13.7%	775
<i>Quintile 3 Income</i>	71 (19)	7.8%	775
<i>Quintile 4 Income</i>	74 (17)	5.2%	774
<i>Highest Income</i>	76 (17)	2.9%	774
Wealth Level			
<i>Lowest Net Worth</i>	65 (23)	11.5%	775
<i>Quintile 2 NW</i>	69 (21)	11.7%	775
<i>Quintile 3 NW</i>	73 (18)	8.4%	775
<i>Quintile 4 NW</i>	75 (17)	5.9%	774
<i>Highest Net Worth</i>	79 (15)	4.4%	774

Table 3 CFM Regressions for Financial Literacy Scores and Age

The first column presents the results of a univariate regression financial literacy score (% correct out of 16 questions – dependent variable) on continuous age (independent variable). The second column presents the same regression using 5-year age categories. The third (age continuous) and fourth (age categorical) columns present multivariate regressions that include a matrix of control variables.

	<i>Linear Age</i>	<i>Cohort Age</i>	<i>Linear Age</i>	<i>Cohort Age</i>
<i>Age</i>	-1.36*** (0.05)		-1.02*** (0.05)	
<i>Age 65-69</i>		-1.62 (1.14)		-1.41 (1.01)
<i>Age 70-74</i>		-9.85*** (1.22)		-5.98*** (1.10)
<i>Age 75-79</i>		-17.36*** (1.33)		-11.8*** (1.22)
<i>Age 80-84</i>		-26.66*** (1.44)		-19.95*** (1.31)
<i>Age 85-89</i>		-32.55*** (1.87)		-25.41*** (1.71)
<i>Age 90+</i>		-41.72*** (3.26)		-32.15*** (3.14)
<i><High School</i>			-10.09*** (1.79)	-10.23*** (1.79)
<i>Some College</i>			7.92*** (1.03)	8.13*** (1.03)
<i>College</i>			13.85*** (1.08)	14.11*** (1.08)
<i>Graduate</i>			16.83*** (1.14)	16.93*** (1.14)
<i>High Income</i>			2.38* (1.20)	2.72** (1.19)
<i>High Wealth</i>			5.47*** (0.99)	5.08*** (0.99)
<i>White</i>			8.82*** (1.15)	8.86*** (1.15)
<i>Male</i>			4.35*** (0.76)	4.28*** (0.76)
<i>Homeowner</i>			4.37*** (1.26)	4.08*** (1.26)
<i>Married</i>			3.96*** (0.80)	3.76*** (0.80)
<i>Tax Sheltered</i>			5.42*** (0.87)	5.32*** (0.87)
<i>Stock/MF</i>			5.43*** (0.84)	5.77*** (0.84)
<i>Sample Size</i>	4152	4152	3898	3898
<i>Adjusted R²</i>	0.17	0.17	0.37	0.37

***, **, * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively

Table 4 CFM Censored Regressions of Financial Literacy and Age

Each column of this table presents the results of a multivariate regression of financial literacy (dependent variable) on age (independent variable) and control variables. The sample is censored by those with a college degree (first column), males only (second column), stockowners (third column), and the older cohort of respondents 60 and older (fourth column).

	<i>College</i>	<i>Males</i>	<i>Stock/MF Owners</i>	<i>Older Cohort</i>
<i>Age</i>	-1.1*** (0.06)	-0.96*** (0.07)	-0.98*** (0.08)	-1.37*** (0.22)
<i><High School</i>		-11.19*** (3.25)	-6.79 (4.89)	-5.53 (3.39)
<i>Some College</i>		8.52*** (1.72)	7.54*** (1.81)	5.00** (2.12)
<i>College</i>		15.95*** (1.73)	11.89*** (1.77)	11.04*** (2.31)
<i>Graduate</i>		17.22*** (1.75)	15.37*** (1.78)	12.7*** (2.54)
<i>High Income</i>	3.53** (1.34)	2.25 (1.68)	2.64 (1.62)	4.29 (4.13)
<i>High Wealth</i>	5.3*** (1.09)	4.84*** (1.35)	5.35*** (1.27)	0.77 (2.43)
<i>White</i>	8.75*** (1.42)	8.36*** (1.75)	6.82*** (1.89)	5.47** (2.56)
<i>Male</i>	5.4*** (0.9)		4.15*** (1.15)	6.76*** (1.72)
<i>Homeowner</i>	4.67*** (1.67)	6.84*** (2.21)	1.16 (2.82)	-1.14 (2.25)
<i>Married</i>	4.37*** (0.97)	6.53*** (1.26)	3.59*** (1.26)	2.44 (1.79)
<i>Tax Sheltered</i>	6.04*** (0.98)	6.16*** (1.27)	4.04*** (1.16)	9.26*** (1.97)
<i>Stock/MF</i>	5.25*** (0.98)	4.08*** (1.26)		7.36*** (1.93)
<i>Sample Size</i>	2,803	1702	1,654	800
<i>Adjusted R²</i>	0.27	0.34	0.23	0.26

***, **, * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively

Table 5 CFM Regressions of Financial Literacy Topic Areas and Age

Each column of this table presents the results of a multivariate regression of individual financial literacy topic areas (dependent variables) on age (dependent variable) and control variables. The first column dependent variable is the four questions on financial literacy basic topics, the second column dependent variable is the four questions on borrowing, the third column dependent variable is the four financial literacy questions on investing, and the fourth column dependent variable is the four insurance-related financial literacy questions.

	Basics	Borrowing	Investment	Insurance
<i>Age</i>	-0.94*** (0.06)	-1.1*** (0.06)	-1.06*** (0.06)	-0.96*** (0.06)
<i><High School</i>	-11.87*** (2.23)	-9.76*** (2.27)	-9.95*** (2.36)	-8.8*** (2.37)
<i>Some College</i>	8.55*** (1.28)	7.07*** (1.31)	8.17*** (1.36)	7.88*** (1.37)
<i>College</i>	15.9*** (1.34)	10.79*** (1.37)	15.72*** (1.42)	12.99*** (1.43)
<i>Graduate</i>	18.78*** (1.42)	13.27*** (1.45)	19.08*** (1.5)	16.18*** (1.51)
<i>High Income</i>	2.76* (1.53)	1.55 (1.49)	3.16** (1.46)	2.05 (1.41)
<i>High Wealth</i>	4.56*** (1.23)	3.94*** (1.28)	8.44*** (1.29)	4.94*** (1.31)
<i>White</i>	11.2*** (1.42)	6.47*** (1.46)	9.99*** (1.51)	7.61*** (1.52)
<i>Male</i>	4.28*** (0.94)	3.95*** (0.96)	6.23*** (0.99)	2.96*** (1.00)
<i>Homeowner</i>	1.1 (1.57)	6.17*** (1.6)	4.82*** (1.66)	5.4*** (1.67)
<i>Married</i>	4.62*** (0.99)	4.11*** (1.01)	3.65*** (1.05)	3.47*** (1.05)
<i>Tax Sheltered</i>	6.81*** (1.08)	3.32*** (1.1)	7.29*** (1.14)	4.25*** (1.15)
<i>Stock/MF</i>	5.89*** (1.03)	3.15*** (1.05)	7.82*** (1.1)	4.87*** (1.11)
<i>Sample Size</i>	3,898	3,898	3,898	3,898
<i>Adjusted R²</i>	0.29	0.23	0.32	0.22

***, **, * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively

Table 6 CFM Logistic Regression Age Coefficients for Individual Financial Literacy Items

This table presents the age coefficients (only) from 16 logistic regressions of each individual financial literacy item (dependent variable) on age (independent variable) and household control variables (education, income, wealth levels, race, gender, marital status and ownership of home, tax sheltered account, and stocks).

<i>Financial Literacy Question</i>	<i>Age Pt. Estimate</i>	<i>St. Beta</i>	<i>Max-rescaled R²</i>
<i>Net worth is equal to:</i>	-5.2%***	0.23 ^c	0.26
<i>If your assets increase by \$5,000 and your liabilities decrease by \$3,000, your net worth would</i>	-5.0%***	0.2 ^a	0.18
<i>Which bank account is likely to pay the highest interest rate on money saved?</i>	-6.2%***	0.28 ^a	0.18
<i>Savings accounts and money market accounts are most appropriate for:</i>	-7.0%***	0.32 ^a	0.20
<i>To reduce the total finance costs paid over the life of an auto loan, you should choose a loan with the</i>	-5.7%***	0.26 ^c	0.28
<i>If you always pay the full balance on your credit card, which of the following is least important?</i>	-6.4%***	0.29 ^a	0.21
<i>On which type of loan is interest never tax deductible?</i>	-7.0%***	0.32 ^a	0.25
<i>Which type of mortgage would allow a first-time home buyer to qualify for the highest loan amount?</i>	-2.9%***	0.13 ^c	0.15
<i>The benefit of owning investments that are diversified is that it</i>	-6.0%***	0.27 ^a	0.19
<i>A young investor willing to take moderate risk for above-average growth would be most interested in:</i>	-4.2%***	0.19 ^a	0.19
<i>The main advantage of a 401(k) plan is that it:</i>	-4.3%***	0.20 ^a	0.12
<i>To ensure that some of your retirement savings will not be subject to income tax upon withdrawal, you would contribute to:</i>	-3.8%***	0.17 ^a	0.11
<i>If you have an insurance policy with a higher deductible, the premiums will be:</i>	-3.9%***	0.17 ^b	0.17
<i>Which of the following types of insurance is most important for single workers without children?</i>	-5.4%***	0.24 ^a	0.22
<i>Which policy provides the most coverage at the lowest cost for a young family?</i>	-4.7%***	0.21 ^a	0.12
<i>Which household would typically have the greatest life insurance need?</i>	-3.9%***	0.17 ^b	0.11

*** indicates significance at the 0.0001 levels.

^{a,b,c,d} indicates age ranks 1st, 2nd, 3rd, & 4th, respectively, of explaining variation in answering question correctly.

Table 7 CFM Financial Confidence Regressions on Age

Each column in this table presents multivariate regression results of financial confidence on age and household control variables. The first column dependent variable is total confidence that combines all of the four confidence items. The second through fifth columns use each of the four confidence items individually as the dependent variable (respondent-reported confidence in managing money, managing credit, using investment products and using insurance products, respectively). The last column presents the logistic regression results of overconfidence (yes/no) as the dependent variable on age (independent variable) along with the matrix of control variables.

	<i>Total Confidence</i>	<i>Managing Money</i>	<i>Managing Credit</i>	<i>Using Investments</i>	<i>Using Insurance</i>	<i>Over- Confident</i>
<i>Age</i>	0.03*** (0.00)	0.01*** (0)	0.02*** (0)	0.04*** (0.01)	0.04*** (0.01)	1.07***
<i>Objective Score (by area)</i>	0.01*** (0.00)	0.00** (0.00)	0.00** (0.00)	0.01*** (0.00)	0.00 (0.00)	
<i><High School</i>	-0.16 (0.18)	0.06 (0.16)	-0.33** (0.17)	-0.14 (0.27)	-0.24 (0.26)	1.48
<i>Some Coll.</i>	-0.11 (0.09)	-0.08 (0.09)	-0.14 (0.09)	0.03 (0.14)	-0.18 (0.14)	0.59***
<i>College</i>	-0.06 (0.10)	-0.09 (0.1)	-0.05 (0.1)	0.2 (0.15)	-0.13 (0.15)	0.45***
<i>Graduate</i>	-0.27*** (0.10)	-0.27*** (0.1)	-0.22** (0.1)	-0.12 (0.16)	-0.32** (0.16)	0.35***
<i>High Income</i>	0.21** (0.09)	0.12 (0.09)	0.14 (0.09)	0.27* (0.14)	0.21 (0.14)	0.69
<i>High Wealth</i>	0.35*** (0.09)	0.26*** (0.09)	0.42*** (0.09)	0.64*** (0.14)	0.18 (0.13)	0.79
<i>White</i>	0.17* (0.10)	0.16 (0.1)	0.38*** (0.1)	0.02 (0.16)	0.13 (0.16)	0.68**
<i>Male</i>	-0.03 (0.07)	-0.13** (0.07)	-0.15** (0.07)	0.28*** (0.1)	-0.13 (0.1)	0.73**
<i>Homeowner</i>	0.47*** (0.12)	0.19* (0.11)	0.36*** (0.12)	0.9*** (0.18)	0.36** (0.17)	1.3
<i>Married</i>	-0.02 (0.07)	-0.02 (0.07)	-0.02 (0.07)	-0.03 (0.11)	0.06 (0.11)	0.9
<i>Tax Sheltered</i>	0.31*** (0.07)	-0.03 (0.08)	0.2*** (0.08)	0.87*** (0.12)	0.2* (0.12)	0.94
<i>Stock/MF</i>	0.38*** (0.07)	0.16** (0.07)	0.33*** (0.07)	0.63*** (0.11)	0.29** (0.11)	0.93
<i>Sample Size</i>	3,455	3,849	3,815	3,590	3,657	3,455
<i>Adjusted R²</i>	0.07	0.01	0.04	0.10	0.01	0.16

***, **, * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively

Table 8 CFM Regressions for Decision Quality on Financial Literacy, Confidence and Age

Each column in this table presents multivariate regression results of decision quality outcomes (dependent variable) on age, financial literacy, financial confidence and age interacted with financial literacy and financial confidence, controlling for household characteristics. The logistic regression dependent variables are having a low mortgage rate (first column), taking advantage of reward cards for convenience credit card users (second column), and the interviewer's assessment of the respondent having a good understanding of the survey questions (third column).

	<i>Low Mortgage Rate</i>	<i>Reward Card</i>	<i>Good Understanding</i>
<i>Age (60-69 reference)</i>			
<i>70-79</i>	0.76	1.86**	1.03
<i>80 and older</i>	0.9	1.84*	1.02
<i>Financial Literacy Score (FinLit)</i>	1.01*	1.02***	1.06***
<i>Overconfident (OC)</i>	0.97	1.58	1.15
<i>FLS*70-79</i>	1.01	1	0.99
<i>FLS*80+</i>	1	1.01	0.98*
<i>OC*70-79</i>	0.87	0.92	1.25
<i>OC*80+</i>	Abnormal value	1.03	0.56
<i><High School</i>	0.44	0.32***	0.78
<i>Some College</i>	0.92	0.99	1.17
<i>College</i>	2.08**	1.37**	1.08
<i>Graduate</i>	1.08	1.59***	1.33
<i>High Income</i>	1.84**	1.05	0.84
<i>High Wealth</i>	2.59***	2.39***	1
<i>White</i>	1.22	1.68***	1.79***
<i>Male</i>	1	0.79**	0.97
<i>Married</i>	1.47*	1.35***	0.9
<i>Sample Size</i>	998	2,511	3,640
<i>R²</i>	0.20	0.30	0.32

***, **, * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively.

Table 9 HRS Regressions of Financial Literacy Scores, Age, and Cognitive Ability

The first column presents the results of a simple univariate regression financial literacy score (% correct out of 7 HRS financial literacy-related questions – dependent variable) on continuous age (independent variable). The second column presents the same regression using age categories. The third (age continuous) and fourth (age categorical) columns present multivariate regressions that include cognitive ability measures (Word Recall and Vocabulary Ability) along control variables.

	<i>Linear Age</i>	<i>Cohort Age</i>	<i>Linear Age</i>	<i>Cohort Age</i>	<i>Linear Age</i>	<i>Cohort Age</i>
<i>Age</i>	-0.41*** (0.08)		-0.35*** (0.08)		-0.24** (0.1)	
<i>Age 65-69</i>		0.31 (2.14)		1.08 (1.92)		1.89 (2.24)
<i>Age 70-74</i>		-2.4 (1.84)		-0.98 (1.65)		-0.21 (1.94)
<i>Age 75-79</i>		-4.51** (2.04)		-2.69 (1.85)		-1.28 (2.12)
<i>Age 80-84</i>		-5.71** (2.67)		-6.32*** (2.39)		-3.64 (2.63)
<i>Age 85-89</i>		-8.21*** (2.93)		-6.7** (2.64)		-2.91 (2.91)
<i>Age 90+</i>		-12.61*** (4.13)		-6.27* (3.75)		-2.93 (3.98)
<i>Word Recall</i>					0.13*** (0.04)	0.14*** (0.05)
<i>Vocabulary Ability</i>					0.13*** (0.04)	0.13*** (0.04)
<i><High School</i>			-8.59*** (1.72)	-8.58*** (1.73)	-6.36*** (1.93)	-6.37*** (1.95)
<i>Some College</i>			2.6* (1.56)	2.9* (1.57)	0.88 (1.79)	0.89 (1.8)
<i>College</i>			10.52*** (1.69)	10.98*** (1.71)	7.4*** (2.01)	7.42*** (2.03)
<i>High Income</i>			0.81 (1.67)	1.37 (1.67)	0.85 (2.01)	0.86 (2.03)
<i>High Wealth</i>			0.27 (1.74)	0.08 (1.76)	0.95 (1.99)	0.91 (2)
<i>White</i>			3.75** (1.59)	3.19** (1.59)	1.44 (1.86)	0.85 (1.83)
<i>Male</i>			5.74*** (1.22)	5.58*** (1.23)	7.21*** (1.42)	7.3*** (1.44)
<i>Homeowner</i>			3.91** (1.59)	3.72** (1.6)	3.1* (1.8)	3.05* (1.81)
<i>Married</i>			-0.12 (1.4)	0.14 (1.41)	-0.22 (1.6)	-0.08 (1.61)
<i>Tax Sheltered</i>			6.42*** (1.39)	6.46*** (1.4)	5.69*** (1.58)	5.79*** (1.6)
<i>Stock/MF</i>			3.25** (1.53)	2.98* (1.54)	3.46** (1.71)	3.4** (1.72)
<i>Sample Size</i>	1109	1109	1108	1108	887	887
<i>Adjusted R²</i>	0.02	0.01	0.23	0.22	0.23	0.23

Figure 1 **Mean and Predicted Financial Literacy Score by Age**

The figure shows mean CFM financial literacy scores for each year of age and predicted financial literacy score using regression estimates when age is specified as age and age-squared, and as 5-year age groups, in a regression model that controls for household characteristics.

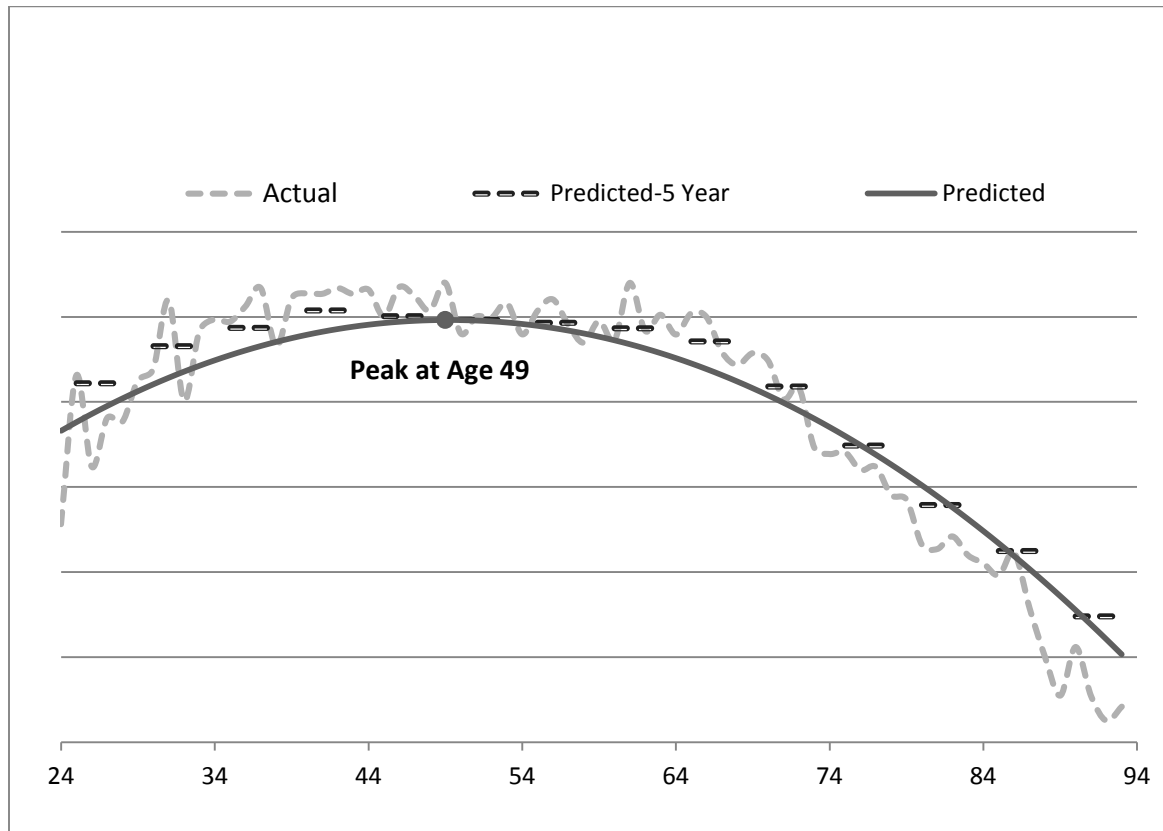


Figure 2 Financial Literacy, Financial Confidence and Cognitive Ability

The figure shows average financial literacy score, average confidence in financial decision making ability, and average percentage of overconfident respondents within each age year using the CFM. Average word recall score within each year of age is drawn from the HRS.

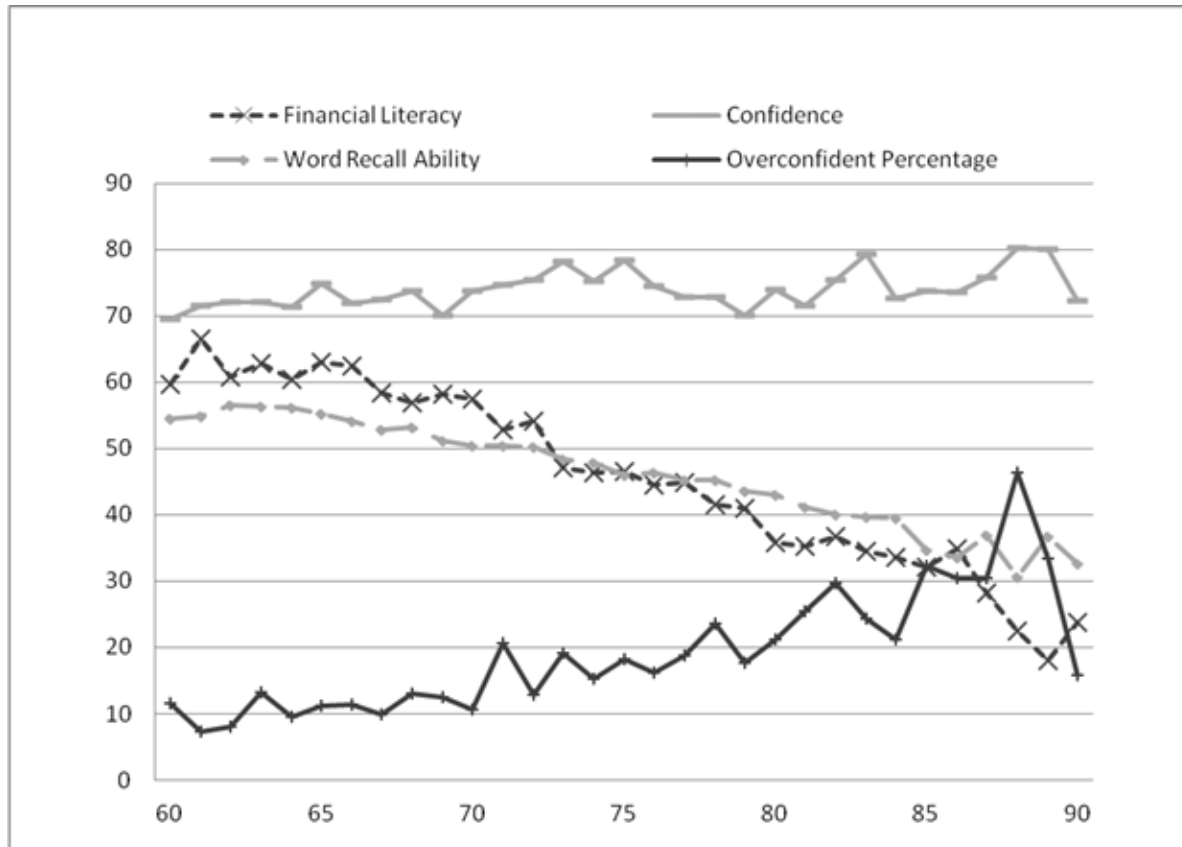
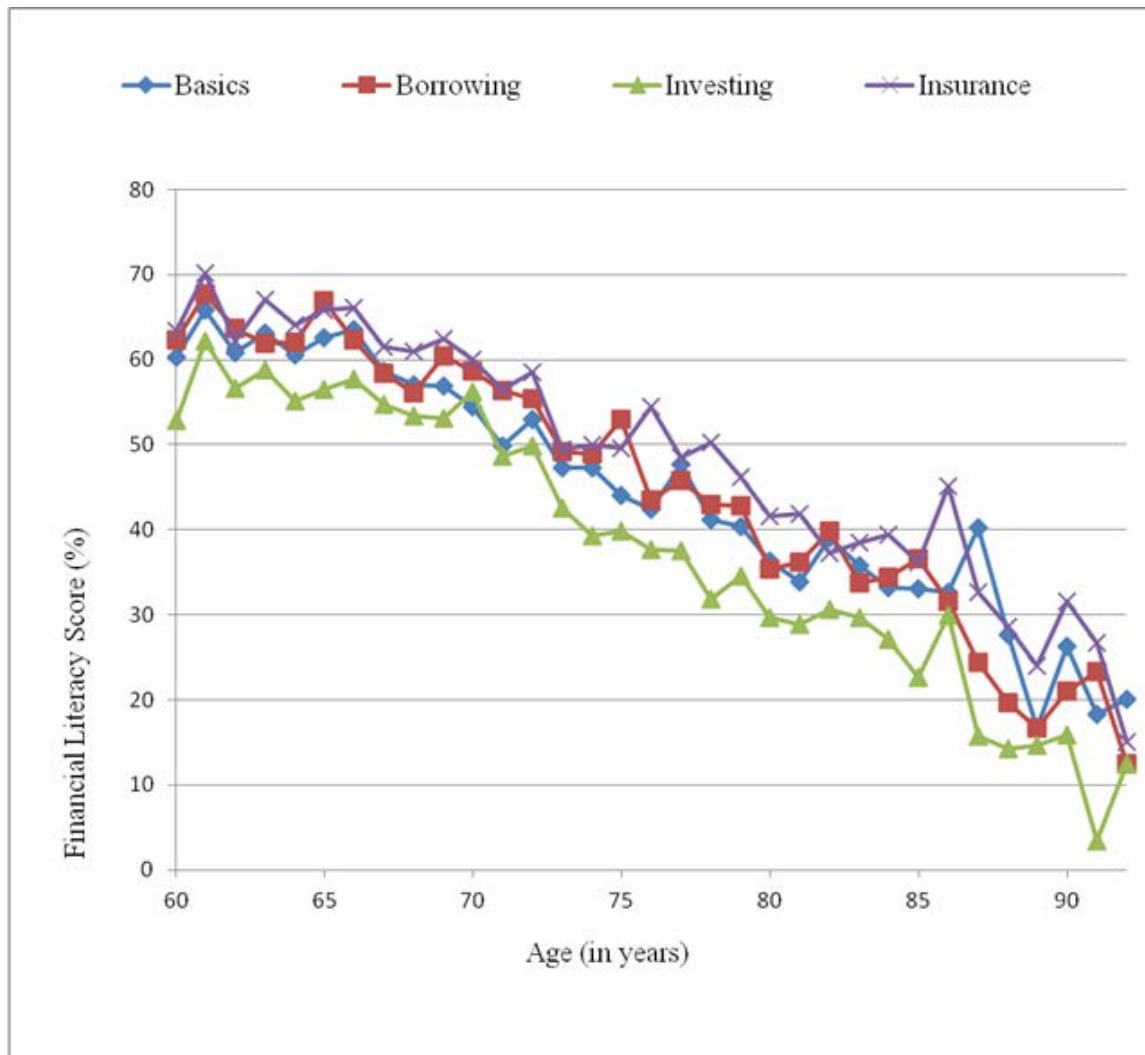


Figure 3 Financial Literacy and Topic Area

The figure shows average financial literacy score within each topic area (basics, borrowing, insurance and investments) by each year of age using the CFM.



Appendix 1-A Financial Literacy Assessment Test (FLAT) Items from the CFM Survey.

Basics Items:

1. Net worth is equal to:
 1. Total assets
 2. *Total assets plus liabilities*
 3. Total assets minus liabilities
2. If your assets increase by \$5,000 and your liabilities decrease by \$3,000, your net worth would
 1. Increase by \$2,000
 2. *Increase by \$8,000*
 3. Increase by \$3,000
3. Which bank account is likely to pay the highest interest rate on money saved?
 1. Savings account
 2. Six month CD or certificate of deposit
 3. *Three year CD*
4. Savings accounts and money market accounts are most appropriate for
 1. Long-term investments like retirement
 2. *Emergency funds and short-term goals*
 3. Earning a high rate of return

Borrowing Items:

5. To reduce the total finance costs paid over the life of an auto loan, you should choose a loan with the
 1. Lowest monthly payment
 2. Longest repayment term
 3. *Shortest repayment*
6. If you always pay the full balance on your credit card, which of the following is least important?
 1. *Annual interest rate*
 2. Annual fees
 3. Line of credit
7. On which type of loan is interest never tax deductible?
 1. A home equity loan
 2. An adjustable rate mortgage
 3. *A personal vehicle loan*
8. Which type of mortgage would allow a first-time home buyer to qualify for the highest loan amount?
 1. Fixed-rate mortgage
 2. *Adjustable-rate mortgage*
 3. Reverse mortgage

Investing Items:

9. The benefit of owning investments that are diversified is that it
 1. *Reduces risk*
 2. Increases return
 3. Reduces tax liability
10. A young investor willing to take moderate risk for above-average growth would be most interested in:
 1. Treasury bills
 2. Money market mutual funds
 3. *Balanced stock funds*

11. The main advantage of a 401(k) plan is that it:
 1. Provides a high rate of return with little risk
 2. *Allows you to shelter retirement savings from taxation*
 3. Provides a well-diversified mix of investment assets
12. To ensure that some of your retirement savings will not be subject to income tax upon withdrawal, you would contribute to:
 1. A Traditional IRA or Individual Retirement Account
 2. *A Roth IRA*
 3. A 401(k) plan

Insurance Items:

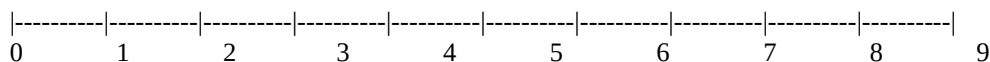
13. If you have an insurance policy with a higher deductible, the premiums will be
 1. Higher
 2. *Lower*
 3. The same
14. Which of the following types of insurance is most important for single workers without children?
 1. Life insurance
 2. *Disability income insurance*
 3. Dental insurance
15. Which policy provides the most coverage at the lowest cost for a young family?
 1. *Renewable term life*
 2. Whole life
 3. Universal life
16. Which household would typically have the greatest life insurance needs?
 1. A middle-class retired couple
 2. A middle-aged working couple with children in college
 3. *A single-earner family with two young children in pre-school*

Confidence Items:

For the following 4 questions, record a number from 0 (LOWEST confidence) to 10 (HIGHEST confidence):

LOWEST CONFIDENCE

HIGHEST CONFIDENCE



10

1. How confident are you with managing money? _____
2. How confident are you with managing credit and debt? _____
3. How confident are you with using investment products? _____
4. How confident are you with using insurance products? _____

Appendix 1-B Financial Literacy-related Items from the HRS Survey.

Question 1.

First, suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow -- more than \$102, exactly \$102, or less than \$102?

1. *More than \$102*
2. Exactly \$102
3. Less than \$102

Question 2.

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, would you be able to buy more than today, exactly the same as today, or less than today with the money in this account?

1. More than today
2. Exactly the same as today
3. *Less than today*

Question 3.

Do you think that the following statement is true or false: buying a single company stock usually provides a safer return than a stock mutual fund?

- 1 True
- 5 *False*

Question 4.

Which asset do you think historically has paid the highest returns over a long time period, say 20 years or more -- savings accounts, bonds, or stocks?

1. Saving accounts
2. Bonds
3. *Stocks*

Question 5.

An employee of a company with publicly traded stock should have a lot of his or her retirement savings in the company's stock.

- 1 True
- 5 *False*

Question 6.

It is best to avoid owning stocks of foreign companies.

- 1 True
- 5 *False*

Question 7.

If the interest rate falls, bond prices will rise.

- 1 *True*
- 5 False